

Lookup Checks. Plookup. Logup

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Distributed Lab

 zkdl-camp.github.io

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Introduction

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In Total: $3n$ constraints for *one XORing operation*. E.g., for 8-bit XORing we would have approximately 24 constraints per operation.

Motivation (cont.)

Suppose you need to compute **10k** 8-bit XOR operations during hashing (that corresponds to $\approx$ four SHA256 512-bit blocks).

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With lookup, you pay 2^{16} constraints to commit to the lookup table, and then you use only 1 constraint per XOR. Thus in total you would have $2^{16} + 10k \approx 75k$ constraints. So we get 3.2 $\times$ boost!

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More generally, lookup check allows to reduce complexity from $O(mn)$ (where m is the cost of each operation and n is the number of operations) to $O(n + d)$, where d is the lookup size.

Application to Bionetta



In *Bionetta*, lookup checks allow to reduce the number of constraints $O(n)$ (where n is the circuit size in Groth16) to sublinear $O(n/\log n)$.

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In *UltraGroth*, by splitting the integer into w -bit limbs, one can reduce the complexity per ReLU down to b/w with the lookup commitment cost of 2^w constraints. Thus, in total, one gets $O(2^w + \ell b/w)$ constraints with ℓ being the number of ReLUs.

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For optimal value of w , this reduces to $O(\ell b/\log(\ell b))$.

Formalizing the problem

Definition

Lookup check consists in proving/verifying that $\{z_i\}_{i \in [n]} \subseteq \{t_j\}_{j \in [d]}$. We will denote multisets as $\vec{z} := \{z_i\}_{i \in [n]}$ and $\vec{t} = \{t_j\}_{j \in [d]}$ and write inclusion check $\vec{z} \subseteq \vec{t}$ for short.

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- $\vec{z} = \{10, 6, 7, 1, 1, 6, 10, 7, 1\} \subseteq \{1, 6, 7, 10\} = \vec{t}$. On the other hand, we have $\vec{z}^* := \{1, 6, 10, 5\} \not\subseteq \vec{t}$.

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- For $\vec{t} = \{t_j\}_{j \in [2^w]}$ with $t_j = j$, condition “ $\vec{z} \subseteq \vec{t}$ ” means “every element z_i is a w -bit integer”.

Working with tuples

In particular, XOR example can also be reduced to this check.

- First initialize all tuples $\vec{t} := \{(a_i, b_i, c_i)\}_{i \in [2^{2n}]}$ such that $a_i \oplus b_i = c_i$ (perceive each a_i, b_i, c_i as a field element from $\mathbb{F}$).

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- We need to check whether $\vec{z} := \{(x_i, y_i, w_i)\}_{i \in [m]} \subseteq \vec{t}$. The verifier samples the challenge $\beta \leftarrow \$ \mathbb{F}$, and lookup table is perceived as $\vec{t}_\beta = \{a_i + \beta b_i + \beta^2 c_i\}_{i \in [2^{2n}]}$, witness as $\vec{z}_\beta := \{x_i + \beta y_i + \beta^2 w_i\}_{i \in [m]}$, and then the check is $\vec{z}_\beta \subseteq \vec{t}_\beta$ as usual.

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Conclusion

Lookup checks are cool, so let us study them!

Plookup

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$$\prod_{i \in [n]} (\gamma + a_i) = \prod_{i \in [n]} (\gamma + b_i), \quad \gamma \leftarrow \$ \mathbb{F}$$

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Can we use it for checking $\{z_i\}_{i \in [n]} \subseteq \{t_j\}_{j \in [d]}$? No, consider:

$$Z(\gamma) = \prod_{i \in [n]} (\gamma + z_i), \quad T(\gamma) = \prod_{j \in [d]} (\gamma + t_j)$$

We can merely state that roots of $Z(\gamma)$ and $T(\gamma)$ are the same.

plookup: problematic solution

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Given $\vec{s} = \{s_i\}_{i \in [n]}$, denote by $\partial \vec{s}$ the **difference set** $\{s_{i+1} - s_i\}_{i \in [n-1]}$ without zero elements. For example, $\partial\{1, 1, 2, 5, 5\} = \{1, 3\}$.

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Example

Suppose $\vec{t} = \{1, 4, 8\}$ and $\vec{s} = \{1, 1, 4, 8, 8, 8\}$. Notice that $\vec{s} \subseteq \vec{t}$ and moreover $\partial \vec{t} = \partial \vec{s} = \{3, 4\}$.

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Problem. Converse is false. Consider $\vec{s}^* := \{1, 5, 5, 5, 8, 8\}$.

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Attempt #2. Construct $\vec{s}$ — sorted concatenation $(\vec{z}, \vec{t})$. First assert permutation check of $\vec{s}$ and $(\vec{z}, \vec{t})$. Then, assert that $\partial \vec{s} = \partial \vec{t}$.

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This is necessary and sufficient condition for $\vec{z} \subseteq \vec{t}$.

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Suppose $\bar{t} = \{1, 4, 8\}$, $\bar{z} = \{1, 1, 4, 8, 8, 8\}$, and $\bar{z}^* = \{1, 5, 5, 5, 8, 8\}$. For $\bar{t}$ and $\bar{z}$ we see that $\bar{s} = \{1, 1, 1, 4, 4, 8, 8, 8, 8\}$ and clearly $\partial\bar{s} = \{3, 4\} = \partial\bar{t}$. As for $\bar{z}^*$, the sorted concatenation is $\bar{s}^* = \{1, 1, 4, 5, 5, 5, 8, 8, 8\}$ and so $\partial\bar{s}^* = \{1, 3\} \neq \partial\bar{t}$.

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Thus our protocol might simply be running two permutation checks: (a) on $\bar{s}$ and $(\bar{z}, \bar{t})$, and (b) on $\partial\bar{s}$ and $\partial\bar{t}$. However, that's still **two** grand products. We can reduce this to a single grand product.

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Necessary and sufficient condition for lookup check is $\partial_\beta \vec{s} = ((1 + \beta)\vec{z}, \partial_\beta \vec{t})$ for randomly chosen $\beta \leftarrow \mathbb{F}$.

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Intuition. Note that for two same consecutive integers s_i, s_{i+1} , instead of zero, one gets $(1 + \beta)s_i$. Other than that, we also have elements of form $t_i + \beta t_{i+1}$, which obviously form $\partial_\beta \vec{t}$. Thus, concatenating all elements of form $(1 + \beta)s_i$ and $\partial_\beta \vec{t}$ gives $\partial_\beta \vec{s}$.

Illustration

Suppose we, again, have the following witness and table:

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The sorted array is $\vec{s} = \{1, 1, 1, 4, 4, 8, 8, 8, 8\}$. Now sample random $\beta \leftarrow \$ \mathbb{F}$. Then we have:

$$\partial_{\beta} \vec{s} = \{1 + \beta, 1 + \beta, 1 + 4\beta, (1 + \beta)4, 4 + 8\beta, (1 + \beta)8, (1 + \beta)8, (1 + \beta)8\}$$

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$$(1 + \beta)\vec{z} = \{1 + \beta, 1 + \beta, (1 + \beta)4, (1 + \beta)8, (1 + \beta)8, (1 + \beta)8\}$$

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Clearly, $\partial_\beta \vec{s} = ((1 + \beta)\vec{z}, \partial_\beta \vec{t})$.

Protocol Specifics

Now, given $t \in \mathbb{F}^d$, $z \in \mathbb{F}^n$, and $s \in \mathbb{F}^{n+d}$, define bi-variate polynomials Z and T as follows:

$$Z(\beta, \gamma) \triangleq (1 + \beta)^n \prod_{i \in [n]} (\gamma + z_i) \prod_{i \in [d-1]} (\gamma(1 + \beta) + t_i + \beta t_{i+1})$$
$$T(\beta, \gamma) \triangleq \prod_{i \in [n+d-1]} (\gamma(1 + \beta) + s_i + \beta s_{i+1})$$

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Theorem

$Z \equiv T$ if and only if $z \subseteq t$ and s is (z, t) sorted by t .

Similarly to permutation check equation $\prod_j (\gamma + a_j) = \prod_j (\gamma + b_j)$, the rest of the protocol is done to ensure that $Z = T$ in the *PlonKish* manner. See lecture notes for concrete details.

Logup

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Definition

For a function $q(X)/r(X)$ from rational function field $\mathbb{F}(X)$, the **formal derivative** is given by

$$\left(\frac{q(X)}{r(X)} \right)' = \frac{q'(X)r(X) - q(X)r'(X)}{r(X)^2}$$

Logarithmic Derivative

Definition

The **logarithmic derivative** of $q(X) \in \mathbb{F}[X]$ is given by the rational function $\text{LogD}[q] = q'(X)/q(X)$.

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The **logarithmic derivative** of $q(X) \in \mathbb{F}[X]$ is given by the rational function $\text{LogD}[q] = q'(X)/q(X)$.

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$\text{logD}[qr] = \text{logD}[q] + \text{logD}[r]$.

Consequently, we can infer:

$$\text{logD} \left[\prod_{j=1}^n (X + a_j) \right] = \sum_{j=1}^n \text{logD} [X + a_j] = \sum_{j=1}^n \frac{1}{X + a_j}$$

Finally something useful

Theorem (On fractional permutation check)

Let $\vec{a} = \{a_i\}_{i \in [n]}$ and $\vec{b} = \{b_i\}_{i \in [n]}$ be two sequences of elements from $\mathbb{F}$. To verify with overwhelming probability whether two multisets are equal, it suffices to check

$$\sum_{j=1}^n \frac{1}{\gamma + a_j} = \sum_{j=1}^n \frac{1}{\gamma + b_j}$$

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Intuition. Take logarithmic derivative from both sides of a permutation check equation $\prod_{j=1}^n (\gamma + a_j) = \prod_{j=1}^n (\gamma + b_j)$ and you are done (see prev. slide). Other direction is a bit trickier to prove, but still trivial enough.

Central Equation

Theorem (On fractional lookup check)

Given two sequences of elements $\{t_i\}_{i \in [d]}$ and $\{z_i\}_{i \in [n]}$, the inclusion check $\{z_i\}_{i \in [n]} \subseteq \{t_i\}_{i \in [d]}$ is satisfied if and only if there exist the set of multiplicities $\{\mu_i\}_{i \in [d]}$ where $\mu_i = \#\{j \in [n] : z_j = t_i\}$ such that:

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Note

This is the central equation used in a large number of studies on its own. What follows is just one variation of how to apply SumCheck using this equation.

Bringing Sum-Check

Suppose the lookup table size is $d = 2^v$ for some v . Then, the table $\vec{t} = \{t_j\}_{j \in [d]}$ can be viewed as a function $t : \{0, 1\}^v \rightarrow \mathbb{F}$.

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Problem: attempting to define $\vec{z} = \{z_i\}_{i \in [n]}$ in a similar manner fails since in practice $n > d$, so the function $\{0, 1\}^v \rightarrow \mathbb{F}$ has a too-small domain. Instead, split $\vec{z}$ into $m = \lceil n/d \rceil$ f-ns $z_1, \dots, z_m : \{0, 1\}^v \rightarrow \mathbb{F}$.

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In other words, we reduced the problem of $\{z_i\}_{i \in [n]} \subseteq \{t_j\}_{j \in [d]}$ to $\bigcup_{i \in [m]} \{z_i(\mathbf{x})\}_{\mathbf{x} \in \{0,1\}^v} \subseteq \{t(\mathbf{x})\}_{\mathbf{x} \in \{0,1\}^v}$.

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Sum-Check equation is as follows:

$$\sum_{\mathbf{x} \in \{0,1\}^v} \sum_{i \in [m]} \frac{1}{\gamma + z_i(\mathbf{x})} = \sum_{\mathbf{x} \in \{0,1\}^v} \frac{\mu(\mathbf{x})}{\gamma + t(\mathbf{x})},$$

where $\mu(\mathbf{x}) = \sum_{i \in [m]} \#\{\mathbf{y} \in \{0, 1\}^v : z_i(\mathbf{y}) = t(\mathbf{x})\}$.

Running Sum-Check. Or Not?

Idea #1: Run Sum-Check on the sum:

$$\zeta(\mathbf{x}) = \sum_{i \in [m]} \frac{1}{\gamma + z_i(\mathbf{x})} - \frac{\mu(\mathbf{x})}{\gamma + t(\mathbf{x})}$$

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$$\zeta(\mathbf{x}) = \underbrace{\frac{\mu(\mathbf{x})}{\gamma + t(\mathbf{x})} - \frac{1}{\gamma + z_0(\mathbf{x})} - \dots - \frac{1}{\gamma + z_{\ell-2}(\mathbf{x})}}_{\zeta_0(\mathbf{x}), \ell \text{ terms}}$$

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We form $k \approx m/\ell$ helper columns $\{h_i(\mathbf{x})\}_{i \in [k]}$ that satisfy: (a) $h_i(\mathbf{x})$ agrees with $\zeta_i(\mathbf{x})$ over $\{0, 1\}^v$, (b) $\sum_{\mathbf{x} \in \{0, 1\}^v} \sum_{i=1}^k h_i(\mathbf{x}) = 0$.

Enforcing correct helper columns

Idea #2: Combine k zero-checks using random scalars $\{\beta_i\}_{i \in [k]} \leftarrow \mathbb{F}$ and merge into a single Sum-Check protocol.

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Example

For $\ell = 2$, $h_0(\mathbf{x}) = \frac{\mu(\mathbf{x})}{\gamma + t(\mathbf{x})} - \frac{1}{\gamma + z_0(\mathbf{x})} = \frac{\mu(\mathbf{x})(\gamma + z_0(\mathbf{x})) - \gamma - t(\mathbf{x})}{(\gamma + t(\mathbf{x}))(\gamma + z_0(\mathbf{x}))}$. Thus, we enforce the following equality:

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$$h_0(\mathbf{x})(\gamma + t(\mathbf{x}))(\gamma + z_0(\mathbf{x})) = \mu(\mathbf{x})(\gamma + z_0(\mathbf{x})) - \gamma - t(\mathbf{x})$$

Final Sum-Check

This way, one runs the Sum-Check on the following function for randomly sampled $\mathbf{a} \leftarrow \mathbb{F}^n$:

$$\sum_{r \in [k]} h_r(\mathbf{x}) + \text{eq}(\mathbf{x}; \mathbf{a}) \hat{n}_r \left(h_r(\mathbf{x}) \prod_{i \in I_r} r_i(\mathbf{x}) - \sum_{i \in I_r} q_i(\mathbf{x}) \prod_{j \in I_r \setminus \{i\}} r_j(\mathbf{x}) \right) = 0$$

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- The **first term** enforces that $\sum_{\mathbf{x} \in \{0,1\}^v} \sum_{r \in [k]} h_r(\mathbf{x}) = 0$.

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- The **first term** enforces that $\sum_{\mathbf{x} \in \{0,1\}^v} \sum_{r \in [k]} h_r(\mathbf{x}) = 0$.
- The **second term** checks whether the helper columns are consistent with each $\zeta_i(\mathbf{x})$ (see prev. slide). For this, we check the equality $h_i(\mathbf{x}) \prod_{j \in I_i} r_j(\mathbf{x}) = \sum_{j \in I_i} q_j \prod_{k \in I_i \setminus \{j\}} r_k(\mathbf{x})$ at a random point $\mathbf{a}$ by multiplying by $\text{eq}(\mathbf{x}; \mathbf{a})$ and taking a linear combination of these equations.

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Lemma

The larger ℓ is (size of each chunk), the more complex computations are involved but smaller commitment sizes are required.

Thank you for your attention



zkdl-camp.github.io



github.com/ZKDL-Camp

